

MPS-300 Series DC Power Supply SCPI Protocol Programming Manual

Case Sensitivity

SCPI commands are not case-sensitive: you can use uppercase, lowercase, or any combination of case.

For example: *RST = *rst

*IDN? = *idn?

*RCL = *rcl

Long Form and Short Form

SCPI command keywords can be sent in long form, short form, or a combination of both. This manual provides the long form. The short form is indicated by uppercase characters

For example:

SYSTem:LOCal Long Form

SYST:LOC Short Form

SYSTem:LOC Combination of long and short forms

SYST:LOCal Combination of long and short forms

Note: Each command keyword must be in either long form or short form, and cannot appear in an intermediate mixed form (i.e., part long, part short).

Example: SYSTe:LOCa is illegal; the command will not be executed.

Command Format

According to the command syntax, most commands (and some parameters) are shown in mixed case. Uppercase letters indicate the short form (abbreviation) of the command. For shorter program lines, you can send the abbreviated form.

For better program readability, you can send the long form.

Example: VOLT and VOLTAGE are both acceptable formats. You may use uppercase or lowercase letters. Thus, VOLTAGE, volt, and Volt are all acceptable. Other formats (e.g., VOL and VOLTAG) are invalid and will not be executed.

1. Curly braces { } enclose the parameter options for a given command string. The curly braces are not sent with the command string.
2. Vertical bar | separates multiple parameter choices for a given command string. For example, in the command {0|1|OFF|ON}, you may specify 0, 1, OFF, or ON. The vertical bar is not sent with the command string.
3. Angle brackets < > indicate that you must assign a value to the parameter inside the brackets. For example, VOLTage {<voltage>}. The angle brackets are not sent with the command string. You must supply a value for the parameter, e.g., VOLT 1.23.
4. Colon : is used to separate a command keyword from the next level keyword. Example: SYSTem:LOCal.
5. Question mark ? – By adding a question mark to a command, you can query the current value of the parameter.
Example: MEASure:VOLTage?.
6. Space – You must use a whitespace character, either [TAB] or [SPACE], to separate the parameter from the command keyword.
7. Terminator – The command string sent to the instrument must end with a \r\n (0x0D, 0x0A) character. Terminating the command string always resets the current SCPI command path to the root level.

Remote Interface Connection

The power supply can be connected to the RS-232 interface via the DB9 connector on the rear panel through a level conversion circuit. The following content will help you understand how to control the power supply output via a PC.

Communication Settings

Before performing communication operations, you should first match the power supply to the PC with the following parameters:

Baud rate: 9600

Data bits: 8

Stop bits: 1

Parity: None

IEEE 488.2 Common Commands

***IDN?**

This query command reads the identification string of the power supply.

Return parameters: Manufacturer name, product model, hardware version, software version.

***RST**

This command resets the power supply to the factory default state.

***SAV {<Address >}**

This command saves the current parameters to the specified address (0–9).

Example: *SAV 1

***RCL {<Address >}**

This command recalls the parameters from the specified address (0–9).

Example: *RCL 1

SYSTem Commands

The SYSTem commands are used to set and query system-related status.

SYSTem:LOCal

This command sets the power supply to local operation mode.

SYSTem:REMOte

This command sets the power supply to remote operation mode.

SYSTem: BEEP {0 | 1 | OFF | ON}

This command enables or disables the power supply's beep sound.

Examples:

SYST:BEEP OFF → "Disable the power supply's beep sound"

SYST:BEEP 1 → "Enable the power supply's beep sound"

SYSTem:BEEP?

This command queries the beep status of the power supply.

Example: SYST:BEEP?

Return parameter: 0 (beep disabled) | 1 (beep enabled)

SYSTem:ERR?

This command queries the power supply error information and clears the error flag.

Example: SYST:ERR?

Return parameter: Error message

SYSTem:TEMP?

This command queries the internal temperature of the power supply.

Example: SYST:TEMP?

Return parameter: Temperature

APPLy Commands

The **APPLy** commands are used to set and query the voltage and current setpoints simultaneously.

APPL y {<voltage>,<current>}

This command sets both the output voltage and current setpoints of the power supply at the same time.

Example: APPL 12.345,1.234

Sets the voltage to 12.345 V and the current to 1.234 A.

APPLy?

This command queries the present voltage and current setpoints simultaneously.

Example: APPL?

Return parameters: Voltage setpoint (XX.XXX), Current setpoint (X.XXX)

MEASure Commands

The **MEASure** commands are used to query the actual output voltage and current values of the power supply.

MEASure: VOLTage?

This command queries the actual output voltage value of the power supply.

Example: MEAS:VOLT?

Return parameter: Actual voltage value (X.XXX)

MEASure: CURRent?

This command queries the actual output current value of the power supply.

Example: MEAS:CURR?

Return parameter: Actual current value (X.XXX)

MEASure: POWer?

This command queries the actual output power of the power supply.

Example: MEAS:POW?

Return parameter: Actual power value (XX.XXX)

MEASure: VCM?

This command queries the actual output voltage and current values of the power supply.

Example: MEAS:VCM?

Return parameter: Actual voltage and current values (XX.XXX, X.XXXX)

OUTPut Commands

The **OUTPut** commands are used to set and query the power supply's output status and output time.

OUTPut {<0 | 1 | OFF | ON>}

This command enables or disables the power supply output state.

Examples:

OUTP OFF → "Disable power supply output"

OUTP 1 → "Enable power supply output"

OUTPut?

This command queries the power supply output state.

Example: OUTP?

VOLTage Commands

The VOLTage commands are used to set and query the voltage setpoint and overvoltage protection (OVP) value.

VOLTage {<voltage>}

This command sets the output voltage setpoint of the power supply.

Example: VOLT 12.345 → Sets the voltage to 12.345 V

VOLTage?

This command queries the voltage setpoint of the power supply.

Example: VOLT?

Return parameter: Voltage setpoint (X.XXX)

VOLTage?MAX

This command queries the maximum output voltage value of the power supply.

Example: VOLT? MAX

Return parameter: Maximum voltage value (XX.XXX)

VOLTage: PROTection {<ovp>}

This command sets the overvoltage protection (OVP) value of the power supply.

Example: VOLT:PROT 12.3 → Sets the OVP value to 12.3 V

VOLTage: PROTection?

This command queries the overvoltage protection value of the power supply.

Example: VOLT:PROT?

Return parameter: OVP value (X.XXX)

VOLTage: PROTection:STAT{<0|1|ON|OFF>}

This command sets the overvoltage protection status of the power supply.

Example: VOLT:PROT:STAT OFF → Sets the OVP status to disabled

VOLTage: PROTection:STAT?

This command queries the overvoltage protection status of the power supply.

Example: VOLT:PROT:STAT?

Return parameter: 0 (disabled) | 1 (enabled)

CURRent Commands

The CURRent commands are used to set and query the current setpoint and overcurrent protection (OCP) status.

CURRent {<current>}

This command sets the output current setpoint of the power supply.

Example: CURR 2.345 → Sets the output current setpoint to 2.345 A

CURRent?

This command queries the current setpoint of the power supply.

Example: CURR?

Return parameter: Current setpoint (X.XXXX)

CURRent?MAX

This command queries the maximum output current value of the power supply.

Example: CURR? MAX

Return parameter: Maximum current value (X.XXXX)

CURRent: PROTection {<ocp>}

This command sets the overcurrent protection (OCP) value of the power supply.

Example: CURR:PROT 3 → Sets the OCP value to 3 A

CURRent: PROTection?

This command queries the overcurrent protection value of the power supply.

Example: CURR:PROT?

Return parameter: OCP value (X.XXXX)

CURRent: PROTection:STAT {<0|1|ON|OFF>}

This command sets the overcurrent protection status of the power supply.

Example: CURR:PROT:STAT ON → Enables overcurrent protection

CURRent: PROTection:STAT?

This command queries the overcurrent protection status of the power supply.

Example: CURR:PROT:STAT?

Return parameter: 0 (disabled) | 1 (enabled)